Customer Churn Analysis

What separates customers who leave from those who stay, built as an interactive Tableau dashboard.

Tools Tableau. Calculated fields, parameters, dashboard design

Data 7,043 telecom customers, 26.5% churned

Live dashboard view on Tableau Public

The questions

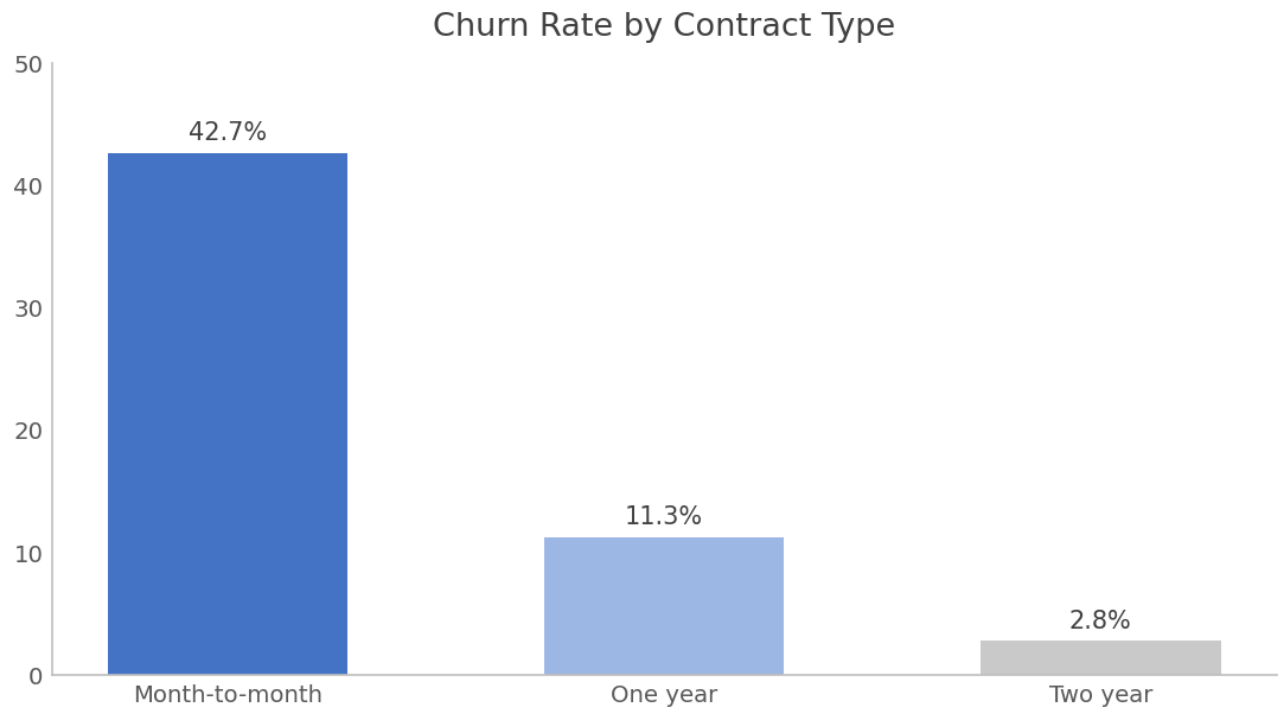
- How much does contract type affect whether a customer churns?
- Does payment method relate to churn?
- How does churn risk change with how long someone has been a customer?
- Do customers who churn pay more or less than customers who stay?

The data

One row per customer: demographics, which services they have, contract type, payment method, monthly and total charges, and whether they churned. A calculated field buckets tenure into 5-month bins so churn rate could be plotted as a trend rather than one point per exact month, and a parameter controls the bin width on the live dashboard.

Contract type is the strongest single factor

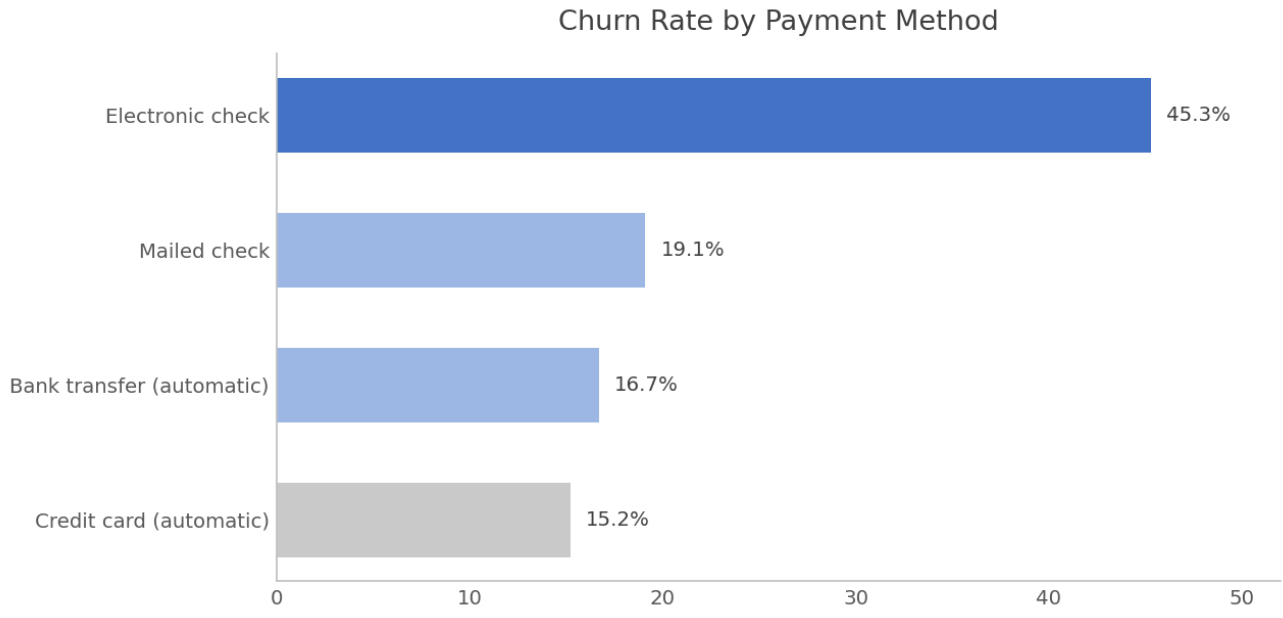
Month-to-month customers churn at 42.7%. One-year contracts drop to 11.3%, and two-year contracts to 2.8%, a 15-fold difference between the shortest and longest commitment. Whatever a customer is annoyed about, a longer contract gives them less opportunity to act on it, and that alone explains more of the variation than any other field in the dataset.



Churn rate by contract length.

Payment method tracks the same divide

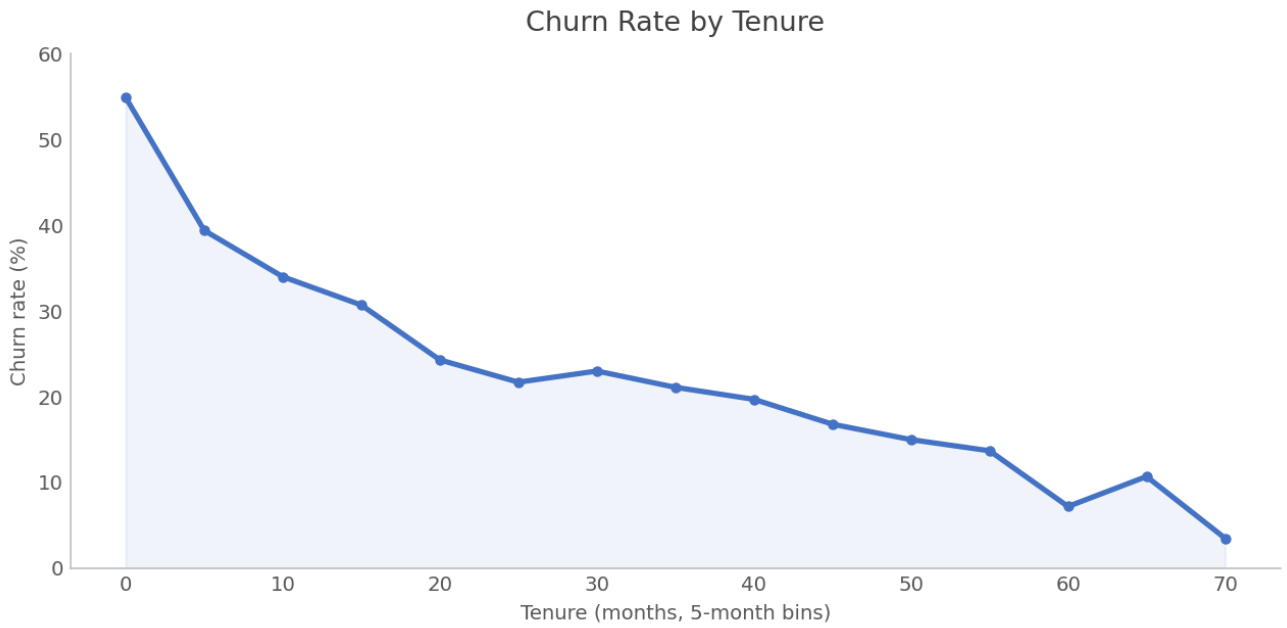
Electronic check carries by far the highest churn rate at 45.3%, roughly three times the other three methods, which all sit between 15 and 19%. Electronic check is also the only payment method not set up as an automatic, recurring charge, so this likely isn't really a payment-method effect, it is the same month-to-month, low-commitment pattern showing up a second time.



Churn rate by payment method.

Risk is highest in the first year and falls steadily

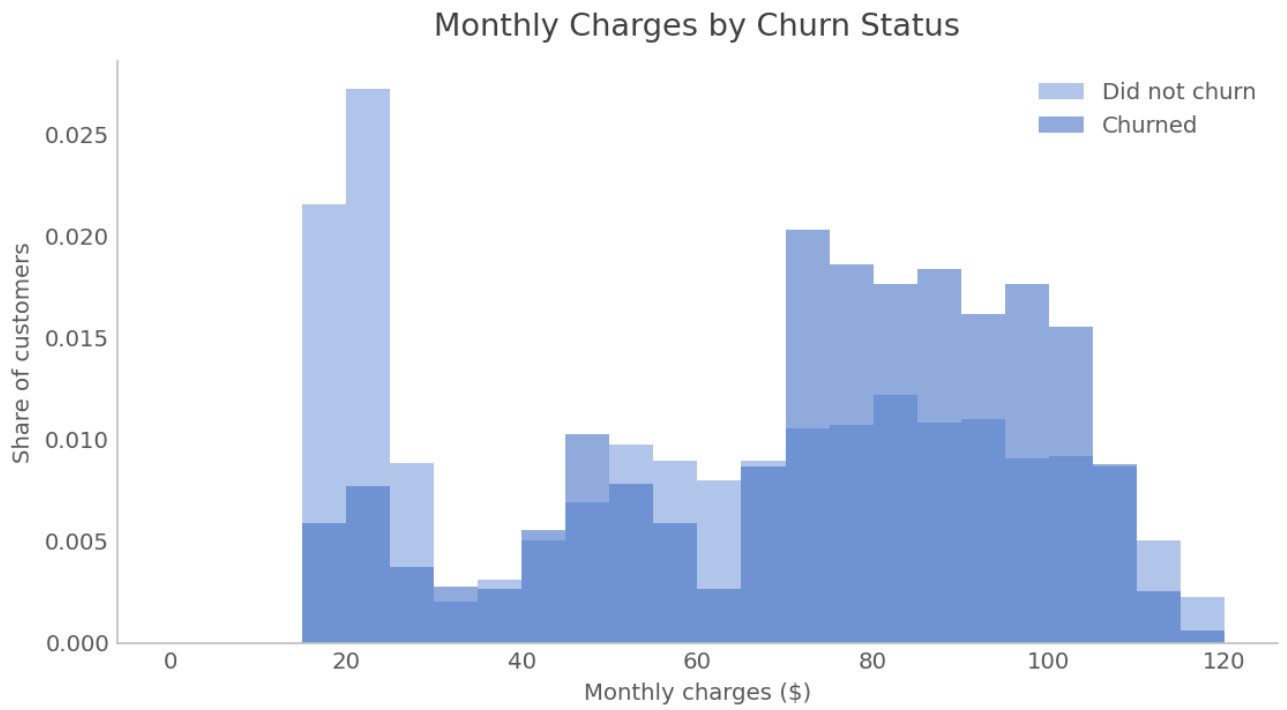
Customers in their first 5 months churn at 54.9%. That falls to 24.3% by the 20-month mark and 3.5% past 70 months. The decline is steady rather than a single cliff edge, which means there is no one obvious moment to intervene. It does mean the first few months carry most of the risk, so that is where retention effort has the most to work with.



Churn rate by tenure, in 5-month bins.

Customers who churn pay more per month, but less in total

Customers who churned paid $74.40 a month on average against $61.30 for those who stayed, but their average total charges were lower, $1,532 against $2,555, because they didn’t stay long enough to accumulate the larger figure. Read alongside the tenure finding, this reads less as "higher bills cause churn" and more as newer, pricier plans attracting customers who have not yet committed.



Distribution of monthly charges, churned versus retained customers.

Interpretation and limitations

The sharpest lever is moving month-to-month customers onto a one-year contract. Moving even a third of the 3,875 month-to-month customers to the one-year rate would be expected to cut several hundred of the roughly 1,655 churns that group produces annually, far more impact than any change aimed at electronic-check users or new customers specifically, since those groups mostly overlap with month-to-month already.

This is a cross-sectional snapshot, not a sequence of events, so it cannot say in which order these factors act, only that they move together. Contract type, payment method and tenure are also correlated with each other, since month-to-month customers are disproportionately new, so some of what looks like three separate effects is likely one underlying pattern seen from three angles.

What is in the workbook

Sheet	Contents
Contract Type Churn	Churn rate by contract length
Payment Method	Churn rate by payment method
Tenure Churn Count	Churn by tenure bucket, with an adjustable bin-width parameter
Total vs Monthly Charges	Charge comparison between churned and retained customers

Portfolio project built on the public Telco Customer Churn dataset.